

THE INDIFFERENCE THRESHOLD OF A LANGUAGE-MODEL JUDGE TRACKS ITS SYMBOL BUDGET

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Paper under double-blind review

ABSTRACT

An evaluator with finite resolution does not order every pair of options. Two options whose consequences differ by less than the evaluator can resolve are indifferent, and the induced preference is a semiorder rather than a weak order. Geometric Evaluation Theory derives that structure from an evaluation object made of a consequence map, a metric with an ideal point, and a resolution budget, and proves that the threshold of the semiorder is set by the budget rather than fitted. We test that prediction on an evaluator the theory did not build. A language-model judge, Qwen2.5-7B-Instruct, is asked to report how far a number lies from a target, and its preference between two options is the order of its reports. Three budgets are set independently and registered before the run. When the numbers the judge is shown, or the symbols it may emit, are placed on a grid, its accuracy at a gap equals the gap divided by the grid step, capped at one, in every cell over four orders of magnitude of step. The step fitted from the judge’s accuracy curve matches the step that was set to within eight percent in every graded cell, with a bootstrap interval that contains it. Quantizing the judge’s weights to 8 bits changes nothing, and quantizing to 4 bits leaves the fitted step unchanged while adding a lapse of three percent above resolution. Resolution and reliability are separate properties of a judge, the first set by the symbols on either side of it and the second by its weights. The registration, its hash, and the graded record are public.

1 INTRODUCTION

A language model used as a judge is asked to tell two things apart. It is shown two answers, or two numbers, or two candidates, and asked which is better or how far each is from what was wanted. The practice is now standard (Zheng et al., 2023), and its known failures are position bias, a preference for round scores, and a dependence on the grading scale (Li et al., 2026). What has not been asked is the question that precedes all of those. How fine a difference can the judge resolve at all, and what sets that limit?

This paper answers the question for one judge on one task, under a theory that says what the answer has to be. The theory is Geometric Evaluation Theory. It begins one step before a preference ordering. An evaluator maps actions into a space of consequences, measures them with a metric of its own from an ideal point of its own, and does so at a finite resolution. The distinctions it can make are derived first, and preference and choice follow from them. At infinite resolution the induced preference is a weak order with a utility. At finite resolution it is a semiorder in the sense of Luce (1956), an ordering with intransitive indifference, and the theory proves that the threshold of that semiorder is the resolution budget. Luce took the threshold as a primitive. Here it is a property of the evaluator fixed by the object that defines it, and that makes the claim one a measurement can fail.

We ran that measurement under a registration sealed before the data were read. The judge is Qwen2.5-7B-Instruct (Qwen Team, 2025), shown a target and a value and asked how far apart they are. Its consequence map is known exactly, since the consequence of a value is its distance from the target. Three budgets were set independently. The judge was shown the numbers at zero to three decimals, was allowed one to twelve tokens for its report, and was run at bfloat16, 8-bit, and 4-bit

weights. The first two budgets are grids placed on the symbols the judge reads and the symbols it writes. The third is a grid placed on the judge itself.

Three things were found. A grid of step h on either side of the judge makes its accuracy at a gap Δ equal to $\min(1, \Delta/h)$, the law of an exact evaluator behind a rounding channel, at every ladder point in every cell, over steps from 0.001 to 1. The step fitted from each accuracy curve matches the step that was set to within eight percent, with a bootstrap interval containing it in every graded cell. Quantizing the weights to 8 bits changed nothing, and quantizing to 4 bits left the fitted step where it was and added a lapse of three percent above resolution. So the judge's resolution is the budget of symbols it is shown and allowed to emit, and its reliability above that resolution is a separate property, which its weights can degrade without touching the resolution.

The paper also reports the theory's identification theorem tested on synthetic evaluators under a sealed registration, where it passed in every cell, and, in the appendix, two further sealed gates, one that held as a population tendency and one that returned indeterminate. The registrations, their hashes, and the graded records are provided as supplementary material.

2 THE EVALUATION OBJECT

Definition 1 (Evaluation object). An evaluation object is a six-tuple $\mathcal{E} = (X, A, C, G, B, \mathcal{K})$ in which X is a set of states carrying a probability law μ , the evaluator's belief; A is a set of actions; $C : A \times X \rightarrow \mathcal{Y}$ is the evaluator, a map into a consequence space $\mathcal{Y}$ that is a C^1 manifold; G is the metric of evaluation, a continuous Riemannian metric on $\mathcal{Y}$ together with an ideal point $t \in \mathcal{Y}$; B is the resolution budget, one of a rank cap k , a length $\varepsilon \geq 0$, or a rate R in bits, with $B = \infty$ for the unbudgeted case; and $\mathcal{K} \subseteq A$ is the set of admissible actions.

The world supplies (X, A, C) and the evaluator supplies $(G, B, \mathcal{K})$. Two evaluators of the same world differ only in the second triple. The split is the source of the theory's content. If the evaluator could also choose C , every weak order on a finite A would be an evaluation object, since a one-dimensional consequence $C(a) = t - d(a)$ represents any real function d . With C fixed by the world, which preferences an evaluator can hold becomes a representation theorem and what its choices reveal becomes a uniqueness theorem.

The evaluation distance of an action is the expected geodesic distance of its consequence from the ideal, $d(a) = (\mathbb{E}_\mu[d_G(C(a, X), t)^2])^{1/2}$, and its cost is $c(a) = d(a)^2$. Under a rank cap k the evaluator retains the top- k eigenspace of the second moment of its habitual consequences and evaluates the projection. Under a length ε the distance is d itself and the budget acts on comparison. Under a rate R the evaluator holds an R -bit description of the consequence and evaluates the reconstruction, and a rate budget is treated as the length budget given by the least attainable distortion. The length budget is the one this paper measures.

3 WHAT THE OBJECT INDUCES

Theorem 1 (Induced distinctions). *Let $\mathcal{E}$ be an evaluation object. (a) The relation $a \sim b$ if and only if $C(a, x) = C(b, x)$ for μ -almost every x is an equivalence relation on A . (b) Under a rank cap k the relation $a \sim_k b$ if and only if the retained projections of $C(a, \cdot)$ and $C(b, \cdot)$ agree almost everywhere is an equivalence relation each of whose classes is a union of classes of $\sim$. (c) Under a length ε the relation $a \approx_\varepsilon b$ if and only if $|d(a) - d(b)| \leq \varepsilon$ is reflexive and symmetric, and it is transitive on A if and only if $\varepsilon = 0$ or the set $\{d(a)\}$ contains no three points $r_1 < r_2 < r_3$ with $r_2 - r_1 \leq \varepsilon$, $r_3 - r_2 \leq \varepsilon$, and $r_3 - r_1 > \varepsilon$.*

Part (c) is the whole difference between a resolution budget and a rank budget. A rank budget coarsens the evaluator's world into classes. A length budget produces a tolerance, and the failure of transitivity is the signature of finite resolution rather than a defect of the evaluator.

Definition 2 (Induced preference). Write d_B for the budgeted distance and ε_B for the length budget, with $\varepsilon_B = 0$ under a rank cap or no budget. Then $a \succ_B b$ if and only if $d_B(a) + \varepsilon_B < d_B(b)$, and $a \sim_B b$ if and only if $|d_B(a) - d_B(b)| \leq \varepsilon_B$.

Theorem 2 (Preference from geometry). *Let A be finite. (a) For $\varepsilon_B = 0$ the relation $\succ_B$ is a weak order represented by the utility $u_B = -d_B$, unique up to strictly increasing transformation. (b) For*

$\varepsilon_B > 0$ the pair $(\succ_B, \sim_B)$ is a semiorder in the sense of Luce, and $u_B = -d_B$ with threshold ε_B is a threshold representation of it. (c) The semiorder of (b) is a weak order if and only if the condition of Theorem 1(c) holds, and as $\varepsilon_B \rightarrow 0$ it converges to the weak order of (a).

Proofs are in Appendix A. Luce (1956) derived semiorders from just-noticeable differences in utility with the threshold as a primitive, Scott & Suppes (1958) gave the representation, and Fishburn (1970) and Tversky (1969) the extensions and the evidence. Here the threshold is the budget and the utility is the distance to the ideal, so both are fixed by the evaluation object. A semiorder with intransitive indifference is what a finite-resolution evaluator must exhibit, and its threshold is a property of the evaluator rather than a fitted constant. Choice from a menu is the set of admissible actions no admissible action strictly beats, nonempty on a finite menu because a semiorder is acyclic, and under a length budget every admissible action within ε of the least distance is chosen, which is satisficing (Simon, 1955) with the aspiration level as the ideal and the tolerance as the budget.

Theorem 3 (Representation, hull law, uniqueness). *Let X be a point, $\mathcal{Y} = \mathbb{R}^m$, and the consequences $y_a = C(a)$ fixed by the world, with the evaluator choosing a positive semidefinite G and an ideal t , so that $c(a) = q(y_a)$ with $q(y) = (y - t)^\top G(y - t)$.*

- (a) *A weak order on finite A is representable by some (G, t) if and only if there is a positive semidefinite $(m + 1) \times (m + 1)$ matrix W such that $Q_W(y) = (y, 1)W(y, 1)^\top$ satisfies $Q_W(y_a) < Q_W(y_b)$ whenever $a \succ b$ and $Q_W(y_a) = Q_W(y_b)$ whenever $a \sim b$. These are linear constraints on W over the semidefinite cone, so representability is a semidefinite feasibility problem.*
- (b) *If a preference is representable, then for every action a and finite set S with y_a in the convex hull of $\{y_s : s \in S\}$ there is $s \in S$ with $a \succsim s$. No action is strictly worse than every action whose consequences surround it, and this holds for every convex evaluation cost.*
- (c) *Let (G_1, t_1) and (G_2, t_2) with $G_1 \neq 0, G_2 \neq 0$ induce the same weak order on a nonempty open connected set U of consequences. Then $G_2 = \lambda G_1$ for some $\lambda > 0$. If in addition $\text{rank } G_1 \geq 2$, or U contains a point at which q_1 attains its minimum, then also $G_1(t_2 - t_1) = 0$.*

Part (b) is the first law of the theory that a scalar utility does not state, and it is checkable on data whenever the consequence map is known. Part (c) says what choices identify. The metric is identified up to scale from any open set of consequences. The ideal is identified up to the metric’s kernel when the metric has rank at least two or the data reach the ideal, and not otherwise: with $\mathcal{Y} = \mathbb{R}, U = (1, 2), q_1(x) = x^2$ and $q_2(x) = (x+10)^2$ the two orders agree on U and $G_1(t_2 - t_1) = -10$. An earlier statement of this theorem omitted the rank or coverage hypothesis and was false in rank one, which an external reader found; the machine-checked version of the result is the whole-space case, which is the case the identification gate of Section 7 uses. A rank budget also produces a reversal that no single utility records: there is an object and two actions with $b \succ a$ at rank cap 1 and $a \succ b$ at rank cap 2, with the same metric and the same ideal at both budgets (Appendix A).

4 A GRID ON THE SYMBOLS IS A LENGTH BUDGET THE EVALUATOR CANNOT SEE PAST

The experiment does not hand the judge a tolerance ε . It places the consequences the judge reads, or the report it writes, on a grid of step h , and the relation this induces on the underlying distances is not the deterministic band of Definition 2. Two distances $0.6h$ apart round to the same value or to different values depending on where they sit relative to the grid. The next proposition says what a grid does to an exact evaluator, and it is the quantity the experiment measures.

Proposition 1 (Grid channel). *Let an evaluator order two options by comparing $\rho(d_1)$ and $\rho(d_2)$, where $d_1 < d_2$ are their true distances and ρ rounds, or truncates, to a grid of step h , and let a tie count as an error. If the position of d_1 relative to the grid is uniform on a grid cell, then the probability that the evaluator orders the pair correctly is*

$$P(\text{correct} \mid \Delta) = \min(1, \Delta/h), \quad \Delta = d_2 - d_1,$$

and the gap at which the evaluator reaches accuracy α is αh . If the evaluator further errs with probability λ independently of the pair, the probability is $(1 - \lambda) \min(1, \Delta/h)$.

Proof. ρ is monotone, so $\rho(d_1) \leq \rho(d_2)$ always, and the pair is ordered correctly exactly when $\rho(d_1) < \rho(d_2)$, that is, when a grid boundary falls in $(d_1, d_2]$. For $\Delta \leq h$ that interval contains a boundary with probability Δ/h under a uniform position, and for $\Delta > h$ it always does. The lapse multiplies by $1 - \lambda$. $\square$

The proposition turns the grid into a length budget in the sense of the theory, with the tolerance read at any accuracy level: the semiorder threshold at level α is αh , and the accuracy curve itself is a straight line through the origin with slope $1/h$. It also fixes what the registered threshold statistic measures. The registration read the threshold as the gap at accuracy 0.9, interpolated linearly in log gap between ladder points. On a ladder with points at $h/2$ and h , where the proposition puts accuracy at 0.5 and 1, that interpolation returns $h \cdot 2^{-0.2} = 0.87h$. The number 0.87 that appears throughout the record is therefore a property of the ladder and the interpolation, not of the judge, and the quantity to report is the fitted step h with its interval. The self-test on a synthetic rounding scorer had already returned $0.87h$ at every h , which is the same fact seen from the other side.

5 THE TEST: A JUDGE WITH A KNOWN CONSEQUENCE MAP

The prediction is that for a fixed world and evaluator the indifference threshold tracks the resolution budget. Coarsening the resolution at which the evaluator resolves consequences enlarges the threshold, and the ordering of pairs separated by more than the largest threshold does not change. The test needs an evaluator whose consequence map is known and whose budget can be set from outside. A language-model judge on a numeric task is such an evaluator, and it is one the theory did not build.

Evaluator. Qwen2.5-7B-Instruct (Qwen Team, 2025) at a recorded revision, loaded in bfloat16, in 8-bit (Dettmers et al., 2022), and in 4-bit NF4 (Dettmers et al., 2023). The judge is a scorer. Shown, in its chat template, a target value and one value, it is asked how far the value is from the target and answers with a number, read as the first number in its greedy generation. The prompt is “Target: {target}. Value: {value}. How far is the value from the target? Answer with a single number and nothing else.” The induced preference between two options compares their reported distances. Equal reports, and any unparsable report, are indifference. This is the theory’s evaluator, a map from a consequence to a scalar cost whose order is the preference, and the model’s own output resolution and arithmetic are its floor.

Two earlier instruments were tried on a probe seed and are kept in the record. A pairwise chooser asked which of two options is closer, read from the next-token logits of the answer letters with both orders shown. With options on either side of the target it was vacuous, choosing the larger number in every straddling pair. With options on one side it reached accuracy 0.99 at a gap of 10 but stayed between 0.43 and 0.63 from gap 0.01 to gap 2, because its first-position preference held the two-order average near one half until the gap was large. A pairwise chooser measures its position bias before it measures a threshold, so it was replaced by the scorer before any pilot.

Consequences. The target is 100. An option is a number in $[50, 150]$ and its consequence is its distance from the target, between 10 and 40, two digits before the decimal point so that a report of fixed length has fixed resolution. A pair is two options whose distances differ by a gap on the ladder 0.0005, 0.001, 0.002, 0.005, 0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 1, 2, 5, 10, 20, each option on either side of the target with equal probability, 200 pairs per gap drawn by seed. The options are drawn continuously, so their positions relative to any grid on the ladder are uniform, which is the hypothesis of Proposition 1.

Budgets. Three, set independently. Rendering precision: the numbers are shown with 0, 1, 2, or 3 decimals, the target at the same precision, so the judge reads consequences rounded to a grid of step 10^{-d} . Report length: the scorer’s answer is limited to k generated tokens, k in 1, 2, 3, 4, 5, 6, 12. This tokenizer emits one character per token for digits and the decimal point (Appendix B), so with two-digit distances a report of k tokens is the true distance truncated to a grid of step 10 at $k = 1$, 1 at $k = 2$ and $k = 3$ (the third character is the decimal point), 0.1 at $k = 4$, 0.01 at $k = 5$, and 0.001 from $k = 6$. This is the length budget taken literally: the evaluator may spend k symbols on its cost, and the grid is the resolution those symbols afford. Weight precision: bfloat16, 8-bit, 4-bit.

Table 1: The step fitted from each accuracy curve against the step that was set, with 95 percent bootstrap intervals, and the lapse above resolution. Rendering cells at 12-token reports; report cells at 3 decimals and bfloat16. The registered threshold at the 0.9 level is shown for the record. Every graded interval contains the set step. The one-token cell is reported and not graded.

Cell	weights	set h	$\hat{h}$	interval	$\hat{h}/h$	lapse $\hat{\lambda}$
0 decimals	bf16, 8-bit, 4-bit	1	0.998	[0.89, 1.14]	1.00	0.000
1 decimal	bf16, 8-bit, 4-bit	0.1	0.0957	[0.086, 0.109]	0.96	0.000
2 decimals	bf16, 8-bit, 4-bit	0.01	0.00924	[0.0083, 0.0104]	0.92	0.000
3 decimals	bf16	0.001	0.00102	[0.00089, 0.00119]	1.02	0.001
3 decimals	8-bit	0.001	0.00102	[0.00089, 0.00118]	1.02	0.001
3 decimals	4-bit	0.001	0.00099	[0.00087, 0.00115]	0.99	0.031 [0.024, 0.038]
1 token	bf16	10	11.9	[10.4, 13.8]	1.19	0.000
2 tokens	bf16	1	1.03	[0.92, 1.17]	1.03	0.000
3 tokens	bf16	1	1.03	[0.92, 1.17]	1.03	0.002
4 tokens	bf16	0.1	0.100	[0.090, 0.113]	1.00	0.001
5 tokens	bf16	0.01	0.0106	[0.0094, 0.0122]	1.06	0.001
6 tokens	bf16	0.001	0.00102	[0.00089, 0.00119]	1.02	0.001

The cells are rendering by weight precision at full report length, twelve cells, and the report ladder at three decimals and full weights, six more.

Registered estimator and bars. For each cell and gap, accuracy is the share of pairs whose preference for the closer option exceeds one half, with ties and unparsable reports counting as one half so that indifference counts against. The registered threshold of a cell is the gap at which accuracy first reaches 0.9, interpolated linearly in log gap between ladder points, infinite if never reached and the smallest gap if reached there. The scorer’s threshold in a cell is predicted to be the larger of the budget it was set and a floor, the scorer’s own resolution at full weights, full rendering, and full report length. P1: along the rendering ladder at fixed weight precision the threshold is non-increasing in decimals within a tolerance factor, and at every decimal count whose step exceeds the floor by a factor of at least 3 it is within that factor of the step. P1b: the same along the report ladder for $k \geq 2$, with the $k = 1$ cell reported and not graded because its resolution of 10 sits at the edge of the gap ladder. P2: along the weight ladder at three decimals and full report length the threshold is non-decreasing as bits are removed, within the factor. P3: for every gap at least twice the largest threshold measured in any cell, accuracy is at least 0.95 in every cell. An anti-vacuity rule required accuracy of at least 0.95 at gap 20 and a floor of at most 0.3 at full precision. The tolerance factor was fixed at 1.5 from a pilot on a separate seed. The registration records that at full weights the rendering and report ladders test only that the scorer’s arithmetic resolves what it is shown and allowed to say, which a competent model makes nearly a tautology, and that the weight ladder is where the prediction can fail.

Channel fit. Proposition 1 was not part of the registration, and the fit that follows is an analysis of the sealed record rather than a registered test. For each cell the step $\hat{h}$ is the least-squares slope of accuracy on gap through the origin over the ladder points below 0.9, and the lapse $\hat{\lambda}$ is one minus the mean accuracy at gaps of at least twice the set step. Intervals are a parametric bootstrap that redraws each gap’s 200 outcomes as binomial and refits, 4000 draws. A paired bootstrap across cells would be preferable and the record does not support it, since it holds per-gap aggregates and not per-pair outcomes.

Discipline. The registration, with its hypothesis, world, statistic, bars, exclusions and null, was committed and its hash recorded before the run seed was drawn. The self-test, the probes, and the pilot were run and recorded before sealing, and the run was graded by a script against the sealed bars.

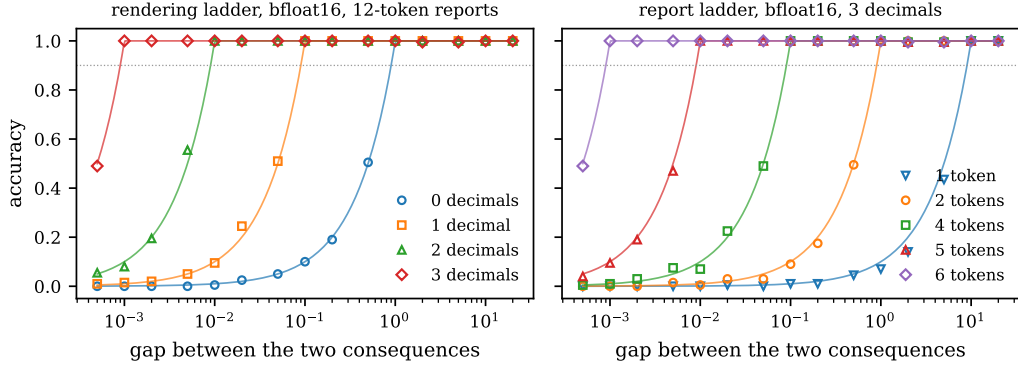


Figure 1: Accuracy against gap, bfloat16 weights, with the grid-channel law $\min(1, \Delta/h)$ drawn through each cell at the step that was set. Left, the rendering ladder at full report length. Right, the report ladder at three decimals. Every point sits on its line: the accuracy at a gap is the gap divided by the grid step, and the line moves by one decade with each step of the budget. The dotted line is the 0.9 level of the registered threshold.

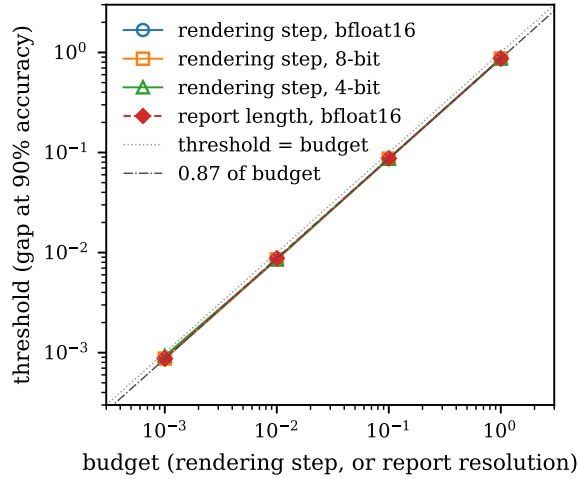


Figure 2: The registered threshold against the budget that was set, log axes. Rendering cells at three weight precisions, and the report ladder. Every point lies on the line at 0.87 of the budget, which is where the registered estimator, log-gap interpolation to the 0.9 level on this ladder, places the threshold of a grid channel. The three weight precisions coincide. The fitted steps in Table 1 are the quantity to read; this figure shows that the sealed statistic behaved as the proposition says it must.

6 RESULTS

The sealed verdict is a pass on P1, P1b, P2 and P3. Figure 1 shows the accuracy curves, Table 1 the fitted steps, and Figure 2 the registered thresholds. No report was unparsable in any cell. The registered thresholds, at bfloat16, were 0.869, 0.0868, 0.00856, and 0.00087 at 0 to 3 decimals and 8.85, 0.872, 0.872, 0.0873, 0.00877, and 0.00087 at 1 to 6 tokens, each with a bootstrap interval of width below three percent of its value, and the 8-bit and 4-bit thresholds equal the bfloat16 values to the digits shown except at three decimals under 4-bit weights, 0.00093, a ratio of 1.06 inside the registered tolerance of 1.5.

The accuracy curve is the grid-channel law. In every cell the accuracy at a gap below the set step equals the gap divided by the step, and above the step it is one, to within the binomial noise of

200 pairs. The largest deviation from $\min(1, \Delta/h)$ at any ladder point in any graded cell is 0.055, at a point where the law predicts 0.5 and the standard error is 0.035. The step fitted from the curve is within eight percent of the step that was set in every graded cell, and every interval contains the set step. That is the content of Figure 1: the curves are one line, Δ/h , drawn at four values of h on the left and five on the right.

Rendering and report length are the same budget. A grid on what the judge reads and a grid on what it writes produce the same accuracy curve at the same step. At three decimals and two tokens the fitted step is 1.03 against 1; at one decimal and full report it is 0.096 against 0.1; at two decimals and five tokens the two fitted steps are 0.0092 and 0.0106 against 0.01. Two and three tokens give the same step because the third character is the decimal point, and six and twelve tokens give the same step because the three-decimal rendering is the floor. This is the length budget of Definition 1 on both sides of the evaluator: the symbols the judge is given and the symbols it may spend set its resolution, and nothing between them does.

Weight precision changes reliability and not resolution. At 8-bit weights every accuracy curve equals its bfloat16 curve to the digits shown. At 4-bit weights the curves at 0, 1, and 2 decimals equal the bfloat16 curves, and at three decimals the fitted step is 0.00099 against 0.00102 at bfloat16, an interval of $[0.00087, 0.00115]$ containing the set step, while the accuracy above resolution falls from 0.99 to 1.00 to a plateau of 0.95 to 0.98. The two-parameter law reads this as a step unchanged and a lapse of 0.031, interval $[0.024, 0.038]$, where the bfloat16 and 8-bit judges have a lapse of 0.001. The registration named the weight ladder as the place where the threshold prediction could fail, and it did not. What the weights did move is a second property, the judge’s reliability once a difference is resolved. On this task, then, a judge has two numbers and not one: a resolution, set by the symbol grid on either side of it, and a reliability above that resolution, which its weight precision can lower without moving the resolution.

Well-separated pairs keep their order, on one gap. The largest registered threshold in any cell is 8.85, in the ungraded one-token cell, so the only ladder gap at least twice that is 20, and P3 as registered is a single-gap test. At gap 20 the worst accuracy over all eighteen cells is 0.985. A budget change enlarged the threshold by four orders of magnitude and reordered nothing at that gap. A registration with gaps beyond 20, and with P3 defined on the largest graded threshold, is the natural next gate and is listed in Section 9.

7 IDENTIFICATION ON SYNTHETIC EVALUATORS

Theorem 3(c) says what choices can recover, and the gate that tests it uses the whole-space case that is machine-checked. Forty evaluators per cell, in two to four dimensions with random metrics including singular ones, were given batteries of consequences of increasing size, and the feasibility program of Theorem 3(a) with a margin was solved from their weak orders alone. Errors were scored against the chance level of guesses from the evaluator prior. In all twelve cells on a fresh seed, under a sealed registration, the metric and the ideal’s range component were recovered at 64 consequences above the dimension, to a few percent of chance, with the error falling at every step as the battery grew, while at or below the design bound, off the span of a battery confined to a subspace, and in the kernel of a singular metric, nothing was recovered. A semiorder that withholds the fifth of pairs closest in distance identified more slowly, to between 5 and 25 percent of chance at the same battery. Two further sealed gates, a hull law on a public survey that held as a population tendency and a shared-representation bound on attention heads that returned indeterminate, are reported in Appendix C at the size of their verdicts.

8 RELATED WORK

Finite resolution producing intransitive indifference is Luce’s semiorder (Luce, 1956), with the representation of Scott & Suppes (1958) and the evidence of Tversky (1969). The present theory adds that the threshold and the budget and the utility is a distance from an ideal, both fixed by the evaluation object, so the semiorder’s parameter is a property of the evaluator rather than a fitted constant. The core of the induced preference, a weighted distance from an ideal point, is Coombs’ unfolding

(Coombs, 1964) and the spatial theory of voting (Davis et al., 1970; Enelow & Hinich, 1984), whose weight matrix is a constant metric of evaluation. Those models are the unbudgeted, single-evaluator case of the object. The nearest neighbor with a budget is rational inattention and limited attention (Sims, 2003; Caplin & Dean, 2015; Masatlioglu et al., 2012; Matějka & McKay, 2015), in which a cost of information about the state produces consideration sets and stochastic choice. The budget here is a different quantity. It limits the resolution of consequences under the evaluator’s metric, not the acquisition of information about states. The decision geometry of Dawid (2007) derives a geometry on distributions from a scoring rule and is the case of an identity evaluator with the true distribution as ideal and no budget. The pullback construction that gives an evaluator its geometry, and the regret bound of Appendix C, come from a companion draft on observer representation (Bond, 2026a), and the economics application is in Bond (2026b).

On language-model judges, Zheng et al. (2023) established the practice and its position and verbosity biases, and Li et al. (2026) found human alignment highest on a 0 to 5 scale, with finer scales worse. That finding is about the scale a judge is asked to use. The present result is about the resolution the judge has, and it says the resolution is the grid on the symbols on both sides of the judge, with the accuracy curve given by Proposition 1. On quantization, Lee et al. (2025) report that 4-bit weights cost small models accuracy on hard tasks and on MT-Bench conversation quality. Our 4-bit judge lost three percent of reliability above resolution and none of its resolution, on a task where the consequence map is known and the resolution is set from outside. The two findings measure different quantities, and the separation of resolution from reliability is what would let them be compared.

9 SCOPE

The result is one model on one numeric task with a known consequence map. At full weights the rendering and report ladders test whether the judge’s arithmetic resolves what it is shown and allowed to say, and a competent model makes that nearly a tautology. The informative parts are Proposition 1 holding at every ladder point rather than only at the threshold, the weight ladder, where the prediction could have failed and instead separated reliability from resolution, and the ordering of the well-separated gap across a four-order-of-magnitude change in threshold. P3 rests on one gap, and the channel fit was not registered. Two things would move this from a result about one judge to a result about judges. The first is a second model family with a different tokenizer, on which the report-length ladder in particular has a different grid. The second is a task that is not literal subtraction while keeping a known consequence map. Both are drafted as the next gate, together with a tokenizer that groups digits, on which the report grid is predicted to follow that tokenizer’s mapping and not this one’s, with gaps of 30, 50, and 100 added so that P3 is a curve, with P3 defined on the largest graded threshold, and with every pair’s reports persisted so that a paired bootstrap is possible. Nothing here is a claim about human evaluators. A human budget manipulation is registered as a later gate and waits on ethics review. What would fail the threshold claim is stated in the registration: a threshold that does not move with resolution where the theory says it must, one that shrinks when resolution is coarsened, or a reordering of well-separated pairs.

AI USE STATEMENT

Generative AI tools were used in this work, and the uses that the ICLR policy lists as requiring disclosure were the following. They assisted in the writing of proofs of the theorems stated here and in the companion draft, in refining the predictions into registered hypotheses with bars, in designing and giving feedback on the experimental protocols, in implementing the methods (the stimulus generators, the judge harness, the estimators, and the grading scripts), in generating the synthetic evaluators on which the identification gate was run, and in interpreting results into the written record. They were not used to generate the theory’s definitions or its central claims, which were the authors’ and precede the tools’ involvement, and they were not used to produce any reported number, which come from executing the registered code on the recorded seeds. Among the uses the policy lists as recommended for disclosure, the tools drafted and edited sections of this paper, searched and summarized related literature, produced the figures from the graded records, wrote and edited code, and formatted references. Verification was as follows. The core theorems are machine-checked in Lean, seven files with no unproven step, with Ky Fan’s principle taken as a named hypothesis; the open-set form of Theorem 3(c) is not among them and was corrected by hand after an external reader found

the defect. Every experiment was registered and sealed with its hash before its run seed was drawn, its self-test and probe were run first, and its verdict was produced by a grading script against the sealed bars rather than by any model. Every citation was checked against the publisher’s record. All AI-assisted text, code, and proofs were reviewed by the authors, who take responsibility for the final content of this work, including text, claims, and artifacts produced with the aid of generative AI.

ETHICS STATEMENT

The survey data used in Appendix C are a public-use release of the American National Election Studies with no identifiers. No human subjects were run. The registered human gate will be submitted for institutional review before any participant is recruited.

REPRODUCIBILITY STATEMENT

The supplementary material contains, for every gate, the sealed registration with its hash, the configuration with model revision and library versions, the seeds, the run log, the graded record, and the grading script that applies the sealed bars, together with the channel-fit script that produces Table 1 and Figure 1 from the graded record. Proofs of every theorem are in Appendix A. The judge is an openly released model at a recorded revision, and the stimulus generator, the scorer prompt, the tokenization of reports, and the estimator are given in full in Section 5, Appendix B, and the supplementary code.

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A PROOFS

Proof of Theorem 1. (a) Equality almost everywhere is an equivalence relation. (b) The relation is equality of a function of a , hence an equivalence, and $C(a, x) = C(b, x)$ implies equality of the retained projections, so each $\sim_k$ class contains the $\sim$ classes of its members. (c) Reflexivity and symmetry are immediate. If three such points exist then $r_1 \approx_\varepsilon r_2$ and $r_2 \approx_\varepsilon r_3$ but not $r_1 \approx_\varepsilon r_3$. If none exist, take $a \approx_\varepsilon b \approx_\varepsilon c$ with values r_a, r_b, r_c . If $r_a \leq r_b \leq r_c$ or the reverse, the absence of such a triple gives $|r_a - r_c| \leq \varepsilon$, and in every other ordering $|r_a - r_c| \leq \max(|r_a - r_b|, |r_b - r_c|) \leq \varepsilon$. $\square$

Proof of Theorem 2. (a) A relation defined by comparing a real function is complete and transitive, and the function represents it. Uniqueness up to monotone transformation is the standard ordinal uniqueness. (b) A relation defined by $u(a) > u(b) + \varepsilon$ with fixed $\varepsilon > 0$ is irreflexive and satisfies the two semiorder axioms. If $a \succ b$ and $c \succ e$, then $u(a) > u(b) + \varepsilon$ and $u(c) > u(e) + \varepsilon$, so either $u(a) > u(e) + \varepsilon$ or $u(c) > u(b) + \varepsilon$, since if both failed we would have $u(a) \leq u(e) + \varepsilon < u(c)$ and $u(c) \leq u(b) + \varepsilon < u(a)$, a contradiction. If $a \succ b \succ c$ and e is any action, then $u(a) > u(c) + 2\varepsilon$, so either $u(a) > u(e) + \varepsilon$ or $u(e) \geq u(a) - \varepsilon > u(c) + \varepsilon$. This is the easy direction of the Scott and Suppes representation of semiorders, and the representation is by construction. (c) The semiorder is a weak order exactly when indifference is transitive, which is Theorem 1(c), and for finite A the condition holds for all sufficiently small ε . $\square$

Proof of Theorem 3. (a) Given (G, t) , put $M = [I \quad -t]$, an $m \times (m+1)$ matrix, and $W = M^\top G M$. Then W is positive semidefinite and $Q_W(y) = (y - t)^\top G (y - t) = q(y)$, so the order constraints hold for W . Conversely let W be positive semidefinite with blocks $G, b/2, b^\top/2, \gamma$, so that $Q_W(y) = y^\top G y + b^\top y + \gamma$. Then G is positive semidefinite, and b lies in the range of G : if $Gz = 0$ then $(z, s)W(z, s)^\top = s b^\top z + \gamma s^2 \geq 0$ for every real s , which forces $b^\top z = 0$. Choose t

with $Gt = -b/2$. Then $Q_W(y) = (y-t)^\top G(y-t) + \gamma - t^\top Gt$ differs from q by a constant, so (G, t) orders A as Q_W does. The order constraints are linear in the entries of W for fixed consequences, and the cone is convex. (b) q is convex because G is positive semidefinite. If $y_a = \sum_s \lambda_s y_s$ with $\lambda_s \geq 0$ summing to one, then $q(y_a) \leq \sum_s \lambda_s q(y_s) \leq \max_s q(y_s)$, so $a \preceq s^*$ for a maximizer s^* . The argument used only convexity. (c) Same weak order on U means $q_2 = \varphi \circ q_1$ on U for a strictly increasing φ on $q_1(U)$, an interval because U is connected and q_1 is continuous.

Metric. Fix $y \in U$ and a direction v with $v^\top G_1 v > 0$, and write $g_i = v^\top G_i v$, $\ell_i = v^\top G_i(y - t_i)$. Along $y + sv$, $q_1 = q_1(y) + 2\ell_1 s + g_1 s^2$ and $q_2 = q_2(y) + 2\ell_2 s + g_2 s^2$, both quadratics in s , and $q_2 = \varphi(q_1)$ on an interval of s around 0. Two quadratics in s related by a strictly increasing map on an interval have the same critical point if the interval contains it, and in general the level sets of q_1 along the segment, which are pairs $\{s, s'\}$ symmetric about $s_1^* = -\ell_1/g_1$, must be level sets of q_2 . If the segment contains two distinct such pairs, q_2 is symmetric about the same point, so $s_2^* = s_1^*$ and φ is affine on the traced interval with slope g_2/g_1 . When rank $G_1 \geq 2$ such a segment exists through every $y \in U$: choose v with $v^\top G_1 v > 0$ and $v^\top G_1(y - t_1) = 0$, which exists because a positive semidefinite form of rank at least two does not vanish on the hyperplane $\{v : v^\top G_1(y - t_1) = 0\}$, and then $s_1^* = 0$ is interior to the segment. When rank $G_1 = 1$, write $G_1 = g w w^\top$; the level sets of q_1 in U are pieces of the hyperplanes $w^\top y = \text{const}$, q_2 is constant on each, so q_2 restricted to U is a function of $w^\top y$ alone, hence a quadratic in $w^\top y$, and its quadratic part is $g' w w^\top$ with $g' > 0$ because q_2 is nonconstant on U . In both cases $G_2 = \lambda G_1$ with $\lambda > 0$: in the rank-one case directly, and in the higher-rank case because φ is affine with the same slope λ on overlapping intervals along every chain of segments through the connected set U , so $q_2 = \lambda q_1 + \mu$ on the open set U , two polynomials that agree on an open set agree identically, and the quadratic parts match.

Ideal. In the case rank $G_1 \geq 2$ the identity $q_2 = \lambda q_1 + \mu$ also matches linear parts, $G_2 t_2 = \lambda G_1 t_1$, which with $G_2 = \lambda G_1$ is $G_1(t_2 - t_1) = 0$. In the case that U contains a minimizer y^* of q_1 , so $G_1(y^* - t_1) = 0$, every segment through y^* in a direction with $v^\top G_1 v > 0$ has $s_1^* = 0$ interior, and the argument above gives $v^\top G_2(y^* - t_2) = 0$ for every such v , hence for a spanning set of the range of G_1 , and with $G_2 = \lambda G_1$ this is $G_1(y^* - t_2) = 0$ and so $G_1(t_2 - t_1) = 0$. When neither hypothesis holds the ideal need not be identified, as the one-dimensional example after the theorem shows. $\square$

Resolution reversal. Take $\mathcal{Y} = \mathbb{R}^2$, $G = I$, $t = 0$, X a point, and a workload whose consequence second moment has top eigenvector e_1 , for example $\text{diag}(4, 1)$. Let $C(a) = (1, 0)$ and $C(b) = (0.9, 2)$. Then $d_1(a) = 1$, $d_1(b) = 0.9$, $d_2(a) = 1$, and $d_2(b) = \sqrt{0.81 + 4} = 2.19$, so $b \succ a$ at rank cap 1 and $a \succ b$ at rank cap 2. A function representing both would satisfy $u(b) > u(a)$ and $u(a) > u(b)$. $\square$

B TOKENIZATION AND PARSING OF REPORTS

The report-length ladder rests on the claim that the judge emits one character per generated token for the digits and the decimal point. Table 2 shows the token identifiers the recorded tokenizer revision assigns to representative reports. Each digit and the decimal point is a single token, so a report limited to k tokens is the decimal expansion of the judge’s answer truncated to k characters. A report is parsed as the first number in the decoded string, a trailing decimal point is dropped, and a string with no number counts as unparsable and as indifference. No report was unparsable in any cell of the run. The probes at full report length, whose raw generations are in the record, show the judge answering with exactly three decimals and rounding the last, for example a true distance of 11.6876 reported as 11.688 and 10.2039 as 10.204. The record does not hold per-pair generations for the run or the count of reports that stopped before the budget, which a future gate will log under the programme’s rule that every stage persists what it saw.

C TWO FURTHER SEALED GATES

Hull law on a public survey, pass as a tendency. Theorem 3(b) needs only a known consequence map and observed choices. In the 1972 American National Election Study respondents placed candidates and parties on seven-point issue scales and rated them on feeling thermometers, so the consequence of an object is its position on a registered pair of scales and the choice is the rating. On

Table 2: Token identifiers for representative reports under the recorded tokenizer revision. One character per token.

report	token identifiers	decoded tokens
10	16, 15	1, 0
10.	16, 15, 13	1, 0, .
10.1	16, 15, 13, 16	1, 0, ., 1
10.12	16, 15, 13, 16, 17	1, 0, ., 1, 2
10.123	16, 15, 13, 16, 17, 18	1, 0, ., 1, 2, 3
27.375	17, 22, 13, 18, 22, 20	2, 7, ., 3, 7, 5
0.001	15, 13, 15, 15, 16	0, ., 0, 0, 1

the registered pair, 489 of 557 respondents had a menu in which the law could bind, and 246 of them, 50.3 percent, violated it at least once, against 66.4 percent when each respondent’s ratings are shuffled among that respondent’s own objects, with none of 200 shuffles reaching the observed rate. Five other pairs of scales gave the same ordering, observed rates of 54 to 63 percent against shuffled rates of 67 to 74 percent. The rates sit well below the geometric base rate of violation and well above zero. On this world the law holds as a population tendency and fails as the deterministic statement of the theorem for about half of the respondents, and the registration records both halves.

Shared-representation regret, indeterminate. Let N evaluators share a representation space $\mathbb{R}^d$ with constant pullbacks $P_1, \dots, P_N$ and a common isotropic source of second moment $\sigma^2 I$, and let a shared representation at budget k be an orthogonal projector Π of rank k . Evaluator i ’s regret from Π is $r_i(\Pi) = \sigma^2(\sum_{j \leq k} \lambda_j(P_i) - \text{tr}(\Pi P_i)) \geq 0$. A shared representation with zero regret for every evaluator exists if and only if there is a k -dimensional subspace that is a top- k eigenspace of every P_i , and for weights $w_i > 0$ the least weighted total regret is attained by a top- k eigenspace of $\bar{P} = \sum_i w_i P_i$, strictly positive exactly when no common eigenspace exists. Both statements are Ky Fan’s maximum principle (Fan, 1949), and the regret formula is the second-order expansion of the loss under a rank- k code. The gate took the three query heads of a grouped-query attention group in a language model as evaluators of one key cache, with shared codes of rank 2, 4, and 8 in 128 dimensions. At those ranks the measured attention loss was 20 to 330 times the second-order prediction, so the regret formula and its bound could not be tested and the verdict is indeterminate. The ordering claim survived on the measured losses alone: the leading eigenspace of the summed geometries beat every random code and the key-covariance code in 32 of 32 cases, while a head’s own leading eigenspace was the best code for that head in fewer than half the cases. A second sealed run perturbed one key at a time by a small isotropic error in the discarded subspace, where the measured loss matched the second-order prediction at the median, a ratio of 1.02 to 1.05, while 64 draws per cell left a 20 percent scatter that missed the registered per-triple bar, so that verdict is indeterminate as well. The registration of the first run also compared a one-key prediction to an all-keys measurement, a defect recorded rather than repaired after the fact.

D PER-CELL ACCURACY

Table 3 gives the accuracy at every ladder gap in every cell of the run, from which the registered thresholds and the fits of Table 1 were computed.

Table 3: Accuracy by gap for all eighteen cells. Columns are the gap ladder. Cells are weight precision, decimals, report tokens.

cell	.0005	.001	.002	.005	.01	.02	.05	.1	.2	.5	1	2	5	10	20
bf16, 0, 12	0.00	0.00	0.00	0.00	0.01	0.03	0.05	0.10	0.19	0.51	1.00	1.00	1.00	1.00	1.00
bf16, 1, 12	0.01	0.01	0.02	0.05	0.10	0.24	0.51	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
bf16, 2, 12	0.06	0.08	0.20	0.56	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
bf16, 3, 12	0.49	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.99	0.99	1.00	1.00
bf16, 3, 1	0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.01	0.01	0.04	0.07	0.14	0.43	1.00	1.00
bf16, 3, 2	0.00	0.00	0.00	0.01	0.01	0.03	0.03	0.09	0.17	0.49	1.00	1.00	1.00	1.00	1.00
bf16, 3, 3	0.00	0.00	0.01	0.02	0.01	0.03	0.03	0.09	0.17	0.49	1.00	0.99	0.99	1.00	1.00
bf16, 3, 4	0.01	0.01	0.03	0.07	0.07	0.23	0.49	1.00	1.00	1.00	1.00	0.99	0.99	1.00	1.00
bf16, 3, 5	0.04	0.10	0.19	0.47	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.99	0.99	1.00	1.00
bf16, 3, 6	0.49	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.99	0.99	1.00	1.00
8-bit, 0, 12	0.00	0.00	0.00	0.00	0.01	0.03	0.05	0.10	0.19	0.51	1.00	1.00	1.00	1.00	1.00
8-bit, 1, 12	0.01	0.01	0.02	0.05	0.10	0.24	0.51	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
8-bit, 2, 12	0.06	0.08	0.20	0.56	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
8-bit, 3, 12	0.49	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.99	0.99	1.00	1.00
4-bit, 0, 12	0.00	0.00	0.00	0.00	0.01	0.03	0.05	0.10	0.19	0.51	1.00	1.00	1.00	1.00	1.00
4-bit, 1, 12	0.01	0.01	0.02	0.05	0.10	0.24	0.51	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
4-bit, 2, 12	0.06	0.08	0.20	0.56	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
4-bit, 3, 12	0.49	0.95	0.97	0.97	0.97	0.95	0.96	0.96	0.98	0.96	0.95	0.98	0.98	0.97	0.98